MBA Fastrack 2025 (CAT + OMETs)

Data Interpretation & Logical Reasoning

DPP: 1

Missing Data

Directions (1-5) Read the following passage and answer the given questions.

The following table represents the marks scored by 4 students, Ashu, Peehu, Kuldeep and Sunny in Eco, Innovation, Finance and Marketing.

	Eco	Innovation	Finance	Marketing
Ashu	40			
Peehu		56		
Kuldeep			45	
Sunny	60			

It is also known that :

1 - The ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

2 - Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

3 - Marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

4 - Total marks scored by these four students is 150 each.

5 - Marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

6 - Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

Q1 What is the ratio of marks scored by Kuldeep and Sunny in Finance respectively ?

(A) 1 : 2 (B) 7 : 8

(C) 5 : 6 (D) 7 : 4

Q2 Marks scored by Kuldeep in Innovation is:

(A) 60 (B) 64

(C) 65 (D) 68

Q3 What is the sum of marks scored by all of them in Marketing ?

(A) 120 (B) 121

(C) 122 (D) 125

Q4 What is the difference in the marks scored by Ashu in Innovation and Marketing ?

(A) 5 (B) 10

(C) 15 (D) 20

Q5 What is the sum of marks scored by Ashu, Peehu, Kuldeep and Sunny in Eco, Innovation, Finance and Marketing respectively?

(A) 95 (B) 105

(C) 120 (D) 135

Directions (6-10) Read the following passage and answer the given questions.

The following table gives data about the amount invested by a person (Principal), rate of interest, and time.

Person	Time (in years)	Rate of Interest	Principal
Anjali	6		14000
Barsha	4		18000
Charu		14%	25500
Divya	2	15%	
Ekta	9	12%	

Note: One has to calculate SI until and unless one is directed to calculate the CI.

Q6 If the ratio of the rate of interest received by Barsha and Charu is 3 : 7 and the ratio of time between Barsha and Charu is 2 : 3 then the difference in the amount received by Charu and Barsha will be.

Q7 If the principal amount invested by Divya and Ekta is same then the ratio of amount received by them will be :

(A) 5 : 8 (B) 5 : 7

(C) 8 : 7 (D) 8 : 11

Q8 If the rate of interest received by Anjali and Barsha is same then the ratio of the amount received by them respectively will be :

(A) 5 : 8

(B) 7 : 6

(C) 2 : 5

(D) Can't be determined

Q9 What is the interest earned by Divya if the principal amount invested by her is 50% less than Charu?

(A) 3800 (B) 3700

(C) 3825 (D) 3870

Q10 If the interest earned by Anjali and Barsha is same then the possible ratio of the rate of interest received by them will be?

(A) 6 : 7 (B) 6 : 11

(C) 6 : 13 (D) 6 : 17

Directions (11-15) Read the following passage and answer the given questions.

The given table shows the number of students in four different departments in four different subjects.

Department	Marketing				Total
	Services	Digital	Luxury	Analytics	
P	-	110	-	150	-
Q	120	-	70	-	400
R	-	130	110	-	480
S	90	140	-	100	-
Total	500	-	-	500	

Q11 If the ratio of the number of students of services marketing from P to R is 18 : 11, find the number of Marketing Analytics from R?

Q12 Total number of Digital marketing students from all the colleges together is 470, then what is the number of Marketing Analytic students from department Q?

Q13 If the ratio of the number of Marketing Analytics students from Q to R is 12 : 13, what is the number of Services Marketing students from R?

Q14 If the ratio of the total number of students from department A and the total number of Marketing Analytics students from all the departments together is 26 : 25. The number of Luxury Marketing from department R is approximately what percent of the total number of students from department A all the subjects together? (only numerical value)

Q15 The number of student in Luxury Marketing from department P is two–third of the students from department S in Luxury Marketing. The total number of students in Luxury Marketing form all departments is 380. Find the number of students in Luxury Marketing form department P?

Directions (16-20) Read the following passage and answer the given questions.

The below table shows the number of three different furniture items (table, chair, and Almirah) manufactured by a company in five weeks.

	Total number of furniture items Manufactured	Percentage of Tables manufactured out of total items	Percentage of Chairs manufactured out of total items	Percentage of Almirah manufactured out of total items
Week 1	300		10	60
Week 2	400	60		20
Week 3		30	20	
Week 4	200		30	40
Week 5		50		20

Note: Some of the data in the above table is missing (shown as blanks).

Q16 If the number of almirahs manufactured in week 3 is 20 more than that of in week 2, what is the respective ratio of the number of tables manufactured in week 3 and week 4 to the number of chairs manufactured in week 1 and week 2?

(A) 12 : 13 (B) 13 : 12

(C) 11 : 13 (D) 12 : 11

Q17 The number of chairs manufactured in week 4 is 20 less than that of in week 3. The number of almirah manufactured in

week 2 and week 4 together is what percentage of the number of tables manufactured in week 1 and week 3 together?
(A) 76.19% (B) 37.49%
(C) 38.39% (D) 95.38%

Q18 If the number of almirahs manufactured in week 5 is 60. What is the difference between the total number of chairs manufactured in week 4 and week 5 together and the total number of tables manufactured in week 2 and week 4 together?
(A) 120 (B) 130
(C) 140 (D) 150

Q19 If the number of tables manufactured in week 5 is 25. The number of almirah manufactured in week 5 is what percentage more or less than the number of chairs manufactured in week 2?
(A) 84.3% (B) 67.2%
(C) 87.5% (D) 73.5%

Q20 What is the average number of tables manufactured by the company in week 1, week 2, and week 4 together?
(A) 120 (B) 130
(C) 140 (D) 150

Directions (21-25) Read the following passage and answer the given questions.
A company manufactured cups of four different types. The table given below shows the number of cups of four different types manufactured by the company in four different weeks of a month.

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	-	-	8820
Type B	-	1520	1650	1730	-
Type C	3240	-	-	2550	8780
Type D	2960	1220	-	-	8270
Total	10730	-	7530	-	-

The number of cups of type D manufactured in week IV is 480 less than the number of cups of type A manufactured in week I. The number of cups of type C manufactured in week II is 1/6 of the total number of the cups of type B manufactured. The number of cups of type A manufactured in week III is 220 more than the number of cups of type B manufactured in week I.

Q21 What is the number of cups of type A manufactured in week IV?
(A) 2570 (B) 2360
(C) 2210 (D) 2430

Q22 What is the ratio of the number of cups of type B manufactured in week I to the number of cups of type C manufactured in week III?
(A) 94 : 93 (B) 87 : 85
(C) 80 : 79 (D) 90 : 87

Q23 Find the total number of cups of all types manufactured in week II.

Q24 What is the number of cups of type D manufactured in week III?
(A) 2110 (B) 2050
(C) 1840 (D) 1920

Q25 What is the absolute difference between the total number of cups of type B manufactured in all the weeks together, and the total number of cups of all types manufactured in week IV?

Directions (26-30) Read the following passage and answer the given questions.
A food delivery company, Swigato, rated five of their delivery employees based on their performance using an integer rating from 0 to 9 for each day of the week (Monday to Sunday). The employees—Girish, Vaibhav, Shikha, Tarun, and Harshit—were assigned alphabetical codes from A to E in that order.
The vertical columns in the table display the mean, median, and mode of ratings for each day.

The horizontal rows provide the sum, median, and mode of the ratings for each employee.

Day	A	B	C	D	E	Mean	Median	Mode
Monday		2				1.4	1	1
Tue		5		2		3.8	5	5
Wed		9			3	5.6	6	9
Thu				0	8	3	1	0
Fri	2		3		9	4.4	4	4
Sat		3		3		1.8	1	1
Sunday			5		6	5.2	6	7
Sum	8	30	28	27	33			
Median	1	4	5	3	5			
Mode	1	-	6	2	1			

Q26 What rating did Girish receive on Wednesday?
[Type '0' if the value cannot be uniquely determined]

1

Q27 What is the sum of Shikha's and Harshit's ratings on Tuesday?
[Type '0' if the value cannot be uniquely determined]

Q28 On which day did none of the delivery employees receive a rating of 1?
(A) Monday
(B) Tuesday
(C) Friday
(D) Cannot be determined

Q29 What is the sum of ratings for employees with code B and D on Thursday?
(A) 8
(B) 2
(C) 6
(D) None of these

Q30 What rating did Vaibhav receive on Sunday?
[Type '0' if the value cannot be uniquely determined]

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Answer Key

Q1	(B)
Q2	(B)
Q3	(C)
Q4	(A)
Q5	(D)
Q6	24600
Q7	(A)
Q8	(D)
Q9	(C)
Q10	(A)
Q11	130
Q12	120
Q13	110
Q14	21
Q15	80
Q16	(D)
Q17	(A)
Q18	(D)
Q19	(C)
Q20	(B)
Q21	(D)
Q22	(A)
Q23	5510
Q24	(D)
Q25	2100
Q26	1
Q27	11
Q28	(C)
Q29	(D)
Q30	7

Hints & Solutions

Note: scan the QR code to watch video solution

Q1. Text Solution:

Here we have some missing data in the given table. One has to find those missing value to solve further.

Here we have given that the ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

But we know marks scored by Peehu in Innovation is 56.

So we can say that Marks scored by Ashu, Kuldeep and Sunny in Innovation will be 40, 64, 48 respectively.

We know that each student score 150 marks in all these four subjects.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep		64		45
Sunny	60	48		

We know that marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

$4x + 3x = (150 - 60 - 48)$

$7x = 42$

$x = 6.$

Marks scored by Sunny in Finance and marketing will be 24 and 18 respectively.

We also know that marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

$20y + 21y = (150 - 64 - 45)$

$41y = 41$

$y = 1$

So, Marks scored by Kuldeep in Eco and Finance will be 20 and 21 respectively.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep	20	64	21	45
Sunny	60	48	24	18

We also know that, Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

So the marks scored by Peehu in Marketing = (150 - 38 - 32 - 56)

= 24.

Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

$z + z = (150 - 80)$

$2z = 70$

$z = 35$

Now the final table will be,

	Eco	Innovation	Finance	Marketing
Ashu	40	40	35	35
Peehu	38	56	32	24
Kuldeep	20	64	21	45
Sunny	60	48	24	18

Required ratio = 21 : 24 = 7 : 8

Video Solution:

Q2. Text Solution:

Here we have some missing data in the given table. One has to find those missing value to solve further.

Here we have given that the ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

But we know marks scored by Peehu in Innovation is 56.

So we can say that Marks scored by Ashu, Kuldeep and Sunny in Innovation will be 40, 64, 48 respectively.

We know that each student score 150 marks in all these four subjects.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing

Ashu	40	40		
Peehu		56		
Kuldeep		64		45
Sunny	60	48		

We know that marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

$4x + 3x = (150 - 60 - 48)$

$7x = 42$

$x = 6.$

Marks scored by Sunny in Finance and marketing will be 24 and 18 respectively.

We also know that marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

$20y + 21y = (150 - 64 - 45)$

$41y = 41$

$y = 1$

So, Marks scored by Kuldeep in Eco and Finance will be 20 and 21 respectively.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep	20	64	21	45
Sunny	60	48	24	18

We also know that, Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

So the marks scored by Peehu in Marketing = (150 - 38 - 32 - 56)

= 24.

Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

$z + z = (150 - 80)$

$2z = 70$

$z = 35$

Now the final table will be,

	Eco	Innovation	Finance	Marketing
Ashu	40	40	35	35
Peehu	38	56	32	24
Kuldeep	20	64	21	45
Sunny	60	48	24	18

Marks scored by Kuldeep in Innovation is 64.

Video Solution:

Q3. Text Solution:

Here we have some missing data in the given table. One has to find those missing value to solve further.

Here we have given that the ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

But we know marks scored by Peehu in Innovation is 56.

So we can say that Marks scored by Ashu, Kuldeep and Sunny in Innovation will be 40, 64, 48 respectively.

We know that each student score 150 marks in all these four subjects.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep		64		45
Sunny	60	48		

We know that marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

$4x + 3x = (150 - 60 - 48)$

$7x = 42$

$x = 6.$

Marks scored by Sunny in Finance and marketing will be 24 and 18 respectively.

We also know that marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

$20y + 21y = (150 - 64 - 45)$

$41y = 41$

$y = 1$

So, Marks scored by Kuldeep in Eco and Finance will be 20 and 21 respectively.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		

Peehu		56		
Kuldeep	20	64	21	45
Sunny	60	48	24	18

We also know that, Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

So the marks scored by Peehu in Marketing = (150 - 38 - 32 - 56)

= 24.

Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

z + z = (150 - 80)

2z = 70

z = 35

Now the final table will be,

	Eco	Innovation	Finance	Marketing
Ashu	40	40	35	35
Peehu	38	56	32	24
Kuldeep	20	64	21	45
Sunny	60	48	24	18

Required sum = 35 + 24 + 45 + 18

= 122

Video Solution:

Q4. Text Solution:

Here we have some missing data in the given table. One has to find those missing value to solve further.

Here we have given that the ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

But we know marks scored by Peehu in Innovation is 56.

So we can say that Marks scored by Ashu, Kuldeep and Sunny in Innovation will be 40, 64, 48 respectively.

We know that each student score 150 marks in all these four subjects.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep		64		45
Sunny	60	48		

We know that marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

4x + 3x = (150 - 60 - 48)

7x = 42

x = 6.

Marks scored by Sunny in Finance and marketing will be 24 and 18 respectively.

We also know that marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

20y + 21y = (150 - 64 - 45)

41y = 41

y = 1

So, Marks scored by Kuldeep in Eco and Finance will be 20 and 21 respectively.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep	20	64	21	45
Sunny	60	48	24	18

We also know that, Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

So the marks scored by Peehu in Marketing = (150 - 38 - 32 - 56)

= 24.

Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

z + z = (150 - 80)

2z = 70

z = 35

Now the final table will be,

	Eco	Innovation	Finance	Marketing
Ashu	40	40	35	35
Peehu	38	56	32	24
Kuldeep	20	64	21	45
Sunny	60	48	24	18

Required difference = 40 - 35 = 5.

Video Solution:

Q5. Text Solution:

Here we have some missing data in the given table. One has to find those missing value to solve further.

Here we have given that the ratio of marks scored by Ashu, Peehu, Kuldeep and Sunny in innovation is 5 : 7 : 8 : 6.

But we know marks scored by Peehu in Innovation is 56.

So we can say that Marks scored by Ashu, Kuldeep and Sunny in Innovation will be 40, 64, 48 respectively.

We know that each student score 150 marks in all these four subjects.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep		64		45
Sunny	60	48		

We know that marks scored by Sunny in Finance and Marketing are in the ratio of 4 : 3.

4x + 3x = (150 - 60 - 48)

7x = 42

x = 6.

Marks scored by Sunny in Finance and marketing will be 24 and 18 respectively.

We also know that marks scored by Kuldeep in Eco and Finance are in the ratio of 20 : 21.

20y + 21y = (150 - 64 - 45)

41y = 41

y = 1

So, Marks scored by Kuldeep in Eco and Finance will be 20 and 21 respectively.

Now the table looks like that,

	Eco	Innovation	Finance	Marketing
Ashu	40	40		
Peehu		56		
Kuldeep	20	64	21	45
Sunny	60	48	24	18

We also know that, Marks scored by Peehu in Eco and Finance are 38 and 32 respectively.

So the marks scored by Peehu in Marketing = (150 - 38 - 32 - 56)

= 24.

Marks scored by Ashu in Finance and Marketing are in the ratio of 1 : 1.

z + z = (150 - 80)

2z = 70

z = 35

Now the final table will be,

	Eco	Innovation	Finance	Marketing
Ashu	40	40	35	35
Peehu	38	56	32	24
Kuldeep	20	64	21	45
Sunny	60	48	24	18

Required sum = 40 + 56 + 21 + 18

= 135.

Video Solution:

Q6. Text Solution:

Time for Charu = $4 \times \frac{3}{2}$ = 6 years

Rate of interest for Barsha = $\frac{3}{7} \times 14 = 6\%$

Amount received by Barsha

$= \frac{18000 \times 4 \times 6}{100} + 18000 = 22320$ Rs.

Amount received by Charu = $25500 + \frac{25500 \times 14 \times 6}{100}$

= Rs. 46920

Required difference = 46920 – 22320

= 24600 Rs.

Video Solution:

Q7. Text Solution:

Let the principal be x.

$$\text{Required ratio} = \frac{x + \frac{x \times 2 \times 15}{100}}{x + \frac{x \times 9 \times 12}{100}}$$

$$\Rightarrow \frac{\frac{130x}{100}}{\frac{208x}{100}} \Rightarrow \frac{130}{208} = \frac{65}{104}$$

$$\Rightarrow 5:8$$

Video Solution:

Q8. Text Solution:

Let the rate be r%

$$\text{Required ratio} = \frac{14000 + \frac{14000 \times r \times 6}{100}}{18000 + \frac{18000 \times r \times 4}{100}}$$

$$\Rightarrow \frac{14000 + 840r}{18000 + 720r}$$

at different value of r we have different ratios.

So, it cannot be uniquely determined.

Video Solution:

Q9. Text Solution:

Principal amount invested by Divya

$$\Rightarrow 50\% \text{ of } 25500$$

$$\Rightarrow 12750$$

$$\text{Interest earned} = 12750 \times \frac{15 \times 2}{100}$$

$$\Rightarrow 3825$$

Video Solution:

Q10. Text Solution:

Interest earned by them are equal.

$$\frac{6 \times R_1 \times 14000}{100} = \frac{4 \times R_2 \times 18000}{100}$$

$$\Rightarrow \frac{R_1}{R_2} \Rightarrow \frac{18000 \times 4}{14000 \times 6}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{9}{7} \times \frac{2}{3} = \frac{6}{7}$$

Video Solution:

Q11. Text Solution:

Total number of students in services marketing = 500

Number of students from P and R = 500 – (120 + 90)

= 500 – 210

= 290

Number of students in services marketing from department R = $\frac{11}{29} \times 2960 = 110$

Number of students in marketing analytics from department R = 480 – (110 + 130 + 110)

= 480 – 350

= 130

Video Solution:

Q12. Text Solution:

Number of students in Digital marketing from Q

= 470 – (110 + 130 + 140)

= 470 – 380

= 90

Number of marketing analytics students from department Q = 400 – (120 + 90 + 70)

= 400 – 280

= 120

Video Solution:

Q13. Text Solution:

Number of marketing analytics students from Q and R = 500 – (150 + 100)

= 500 – 250 = 250

Number of students in marketing analytics from R = $\frac{13}{25} \times 250$ = 130

Number of services marketing students from R = 480 – (130 + 110 + 130)

= 480 – 370

= 110

Video Solution:

Q14. Text Solution:

Total number of students from the department A

$= \frac{26}{25} = \frac{x}{500}$

x = 520

Required percentage = $\frac{110}{520} \times 100$

= 21.15% = 21% (approx.)

Video Solution:

Q15. Text Solution:

Number of students in Luxury marketing from departments P and S = 380 – (70 + 110)

= 380 – 180

= 200

According to the question,

$($ *Student*

2x + 3x = 200

5x = 200

x = 40

2x = 2 × 40

= 80

Video Solution:

Q16. Text Solution:

From the given table:

The number of almirah manufactured in week 3

$=20+\frac{20}{100} \times 400=100$

The number of tables manufactured in week 3 and week 4

$=100 \times \frac{30 \%}{50 \%}+200 \times \frac{30}{100}=120$

The number of chairs manufactured in week 1 and week 2

$=300 \times \frac{10}{100}+400 \times \frac{20}{100}=110$

So, required ratio $=120: 110$

$=12: 11$

Video Solution:

Q17. Text Solution:

From the given table:

The number of chairs manufactured in week 3

$=20+200 \times \frac{30}{100}=80$

The number of almirah manufactured in week 2 and week 4 together

$=400 \times \frac{20}{100}+200 \times \frac{40}{100}=160$

The number of tables manufactured in week 1 and week 3 together

$=300 \times \frac{30}{100}+80 \times \frac{30 \%}{20 \%}=210$

So, required percentage $=\frac{160}{210} \times 100=76.19 \%$

Video Solution:

Q18. Text Solution:

The number of chairs manufactured in week 4 and week 5 together

$=200 \times \frac{30}{100}+60 \times \frac{30 \%}{20 \%}=150$

The number of tables manufactured in week 2 and week 4 together

$$\begin{aligned} & =400 \times \frac{60}{100}+200 \times \frac{30}{100}=300 \\ & \text { Required difference }=300-150=150 . \end{aligned}$$

Video Solution:

Q19. Text Solution:

From the given table:

The number of almirah manufactured in week 5

$=25 \times \frac{20}{50}=10$

The number of chairs manufactured in week 2

$=400 \times \frac{20}{100}=80$

So, required $\%=\frac{80-10}{80} \times 100=87.5 \%$

Video Solution:

Q20. Text Solution:

The number of tables manufactured by the company in week 1

$=300 \times \frac{30}{100}=90$

The number of tables manufactured by the company in week 2

$=400 \times \frac{60}{100}=240$

The number of tables manufactured by the company in week 4

$=200 \times \frac{30}{100}=60$

Required average $=\frac{90+240+60}{3}=130$

Video Solution:

Q21. Text Solution:

The number of cups of type B manufactured in week I = 10730 - 2650 - 3240- 2960 = 1880

The total number of cups of type B manufactured =1880 + 1520 + 1650 + 1730 = 6780

The number of cups of type C manufactured in week II = $6780 \times \frac{1}{6}$ = 1130

The total number of cups of all types manufactured in week II = 1640 + 1520 + 1130 + 1220 = 5510

The number of cups of type A manufactured in week III = 1880 + 220 = 2100

The number of cups of type A manufactured in week IV = 8820 - 2650 - 1640- 2100 = 2430

The number of cups of type D manufactured in week IV = 2650 - 480 = 2170

The number of cups of type C manufactured in week III = 8780 - 3240 - 1130 - 2550 = 1860

The number of cups of type D manufactured in week III = 8270-2960 - 1220 - 2170 = 1920

The total number of cups of all types manufactured in week IV = 2430 +1730 + 2550 +2170 = 8880

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	2100	2430	8820
Type B	1880	1520	1650	1730	6780
Type C	3240	1130	1860	2550	8780
Type D	2960	1220	1920	2170	8270
Total	10730	5510	7530	8880	32650

Required number of cups of type A manufactured in week IV = 2430

Video Solution:

Q22. Text Solution:

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	2100	2430	8820
Type B	1880	1520	1650	1730	6780
Type C	3240	1130	1860	2550	8780
Type D	2960	1220	1920	2170	8270
Total	10730	5510	7530	8880	32650

Required ratio = 1880 : 1860= 94 : 93

Video Solution:

Q23. Text Solution:

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	2100	2430	8820
Type B	1880	1520	1650	1730	6780
Type C	3240	1130	1860	2550	8780
Type D	2960	1220	1920	2170	8270
Total	10730	5510	7530	8880	32650

Required total number of cups of all types manufactured in week II = 5510

Video Solution:

Q24. Text Solution:

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	2100	2430	8820
Type B	1880	1520	1650	1730	6780
Type C	3240	1130	1860	2550	8780
Type D	2960	1220	1920	2170	8270
Total	10730	5510	7530	8880	32650

Required number of cups of type D manufactured in week III = 1920

Video Solution:

Q25. Text Solution:

Types	Week I	Week II	Week III	Week IV	Total
Type A	2650	1640	2100	2430	8820
Type B	1880	1520	1650	1730	6780
Type C	3240	1130	1860	2550	8780
Type D	2960	1220	1920	2170	8270
Total	10730	5510	7530	8880	32650

Required differences = 8880 - 6780 = 2100

Video Solution:

Q26. Text Solution:

To solve this problem, we have to understand that the mode of ratings must have been occurring at least 2 times.

Firstly calculate the sum of ratings from Monday to Sunday, by their mean value.

On monday, mean value is 1.4 so the sum of all values will be 1.4*5 = 7

Similarly for other days,

Day	A	B	C	D	E	Sum	Median	Mode
Monday		2				7	1	1
Tue		5		2		19	5	5
Wed		9			3	28	6	9
Thu				0	8	15	1	0
Fri	2		3		9	22	4	4
Sat		3		3		9	1	1
Sunday			5		6	26	6	7

Sum	8	30	28	27	33			
Median	1	4	5	3	5			
Mode	1	-	6	2	1			

Using Median we can find the middle value among the possible ratings.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday		2				_, _, 1, _, _	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2		3		9	2, 3, 4, _, 9	22	4
Sat		3		3		_, _, 1, 3, 3	9	1
Sunday			5		6	_, _, 6, _, _	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, _, _, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Friday, the sum is 22.

Therefore the missing rating = 22 - (2 + 3 + 4 + 9) = 4, also makes 4 as mode.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday		2				_, _, 1, _, _	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0

Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat		3		3		_, _, 1, 3, 3	9	1
Sunday			5		6	_, _, 6, _, _	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, _, _, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Monday:

The sum = 7

Mode = 1

Known Ratings = 1 and 2

The only possibility for meaning three ratings is 1, 1, and 2

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday		2				1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat		3		3		_, _, 1, 3, 3	9	1
Sunday			5		6	_, _, 6, _, _	26	7
Ratings	_, _, _, 1, _,	_, _, _, 4, _, _, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			

	_, _							
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Saturday:

The sum = 9

Mode = 1

Rating 3 already occurs twice, therefor 1 has to be more than twice.

So remaining ratings is 1 for each person.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday		2				1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday			5		6	_, _, 6, _, _	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, _, _, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Sunday:

The sum = 26

Mode = 7, means 7 must occurs at least twice.

Known Ratings are 5, 6, 7, and 7 making sum of 25

Hence, the remaining rating is 1

Day	A	B	C	D	E	Ratings	Sum	Mode

Monday		2				1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday			5		6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, _, _, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Sunday:

Rating 7 cannot be given to A, will exceed the sum i.e. 8

Hence, A gets 1 and B and D gets 7 each.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday		2				1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9			3	_, _, 6, _, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday	1	7	5	7	6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, 5, 7, 9	_, _, _, 5, _, _, _	0, _, _, 3, _, _, _	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

For D:

Total sum = 27

Median = 3

Mode = 2

Hence 2 occurs twice only.

Known Rating = 2 + 2 + 0 + 4 + 3 + 7 = 18

Remainig rating = 27 - 18 = 9

The only possibility for D to get 9 rating is on Wednesday.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday	1	2	1	2	1	1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed		9		9	3	_, _, 6, 9, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday	1	7	5	7	6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, 5, 7, 9	1, 1, 3, 5, _, _, _	0, 2, 2, 3, 4, 7, 9	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

On Wednesday:

The sum = 28

Knwon Ratings = 9 + 9 + 3 + 6 = 27

Remaining Rating = 28 - 27 = 1

The only possibility for Rating 1 is for A.

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday	1	2	1	2	1	1, 1, 1, 2, 2	7	1
Tue		5		2		_, _, 5, _, _	19	5
Wed	1	9	6	9	3	1, 3, 6, 9, 9	28	9
Thu				0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday	1	7	5	7	6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, 5, 7, 9	1, 1, 3, 5, _, _, _	0, 2, 2, 3, 4, 7, 9	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

For C:

The sum = 28

Mode = 6, means 6 must occurs twice.

Known Ratings = 1 + 1 + 3 + 5 + 6 + 6 = 22

Remaining rating = 28 - 22 = 6

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday	1	2	1	2	1	1, 1, 1, 2, 2	7	1
Tue		5	6	2		_, _, 5, _, _	19	5
Wed	1	9	6	9	3	1, 3, 6, 9, 9	28	9
Thu			6	0	8	0, _, 1, _, _	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday	1	7	5	7	6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _, _, _	_, _, _, 4, 5, 7, 9	1, 1, 3, 5, 6, 6, 6	0, 2, 2, 3, 4, 7, 9	_, _, _, 5, 6, 8, 9			
Sum	8	30	28	27	33			
Mode	1	-	6	2	1			

For B:

The sum = 30

Known Rating = 2 + 5 + 9 + 4 + 3 + 7 = 30

Remaining Rating = 30 - 30 = 0

Day	A	B	C	D	E	Ratings	Sum	Mode
Monday	1	2	1	2	1	1, 1, 1, 2, 2	7	1
Tue		5	6	2		_, _, 5, _, _	19	5
Wed	1	9	6	9	3	1, 3, 6, 9, 9	28	9
Thu		0	6	0	8	0, 0, 1, 6, 8	15	0
Fri	2	4	3	4	9	2, 3, 4, 4, 9	22	4
Sat	1	3	1	3	1	1, 1, 1, 3, 3	9	1
Sunday	1	7	5	7	6	1, 5, 6, 7, 7	26	7
Ratings	_, _, _, 1, _	0, 2, 3, 4, 5, 7, 9	1, 1, 3, 5, 6, 6, 6	0, 2, 2, 3, 4, 7, 9	_, _, _, 5, 6,			

	_, _				8, 9			
Sum	**8**	**30**	**28**	**27**	**33**			
Mode	**1**	**-**	**6**	**2**	**1**			

Hence, on Thursday, A gets Rating 1

Day	**A**	**B**	**C**	**D**	**E**	**Ratings**	**Sum**	**Mode**
Monday	1	2	1	2	1	**1, 1, 1, 2, 2**	**7**	**1**
Tue		5	6	2		**_, _, 5, _, _**	**19**	**5**
Wed	1	9	6	9	3	**1, 3, 6, 9, 9**	**28**	**9**
Thu	1	0	6	0	8	**0, 0, 1, 6, 8**	**15**	**0**
Fri	2	4	3	4	9	**2, 3, 4, 4, 9**	**22**	**4**
Sat	1	3	1	3	1	**1, 1, 1, 3, 3**	**9**	**1**
Sunday	1	7	5	7	6	**1, 5, 6, 7, 7**	**26**	**7**
Ratings	**_, _, _, 1, _, _, _**	**0, 2, 3, 4, 5, 7, 9**	**1, 1, 3, 5, 6, 6, 6**	**0, 2, 2, 3, 4, 7, 9**	**_, _, _, 5, 6, 8, 9**			
Sum	**8**	**30**	**28**	**27**	**33**			
Mode	**1**	**-**	**6**	**2**	**1**			

A's Rating on Tuesday = 8 - (1 + 1 + 1 + 2 + 1 + 1) = 1

Day	**A**	**B**	**C**	**D**	**E**	**Ratings**	**Sum**	**Mode**
Monday	1	2	1	2	1	**1, 1, 1, 2, 2**	**7**	**1**
Tue	1	5	6	2		**_, _, 5, _, _**	**19**	**5**
Wed	1	9	6	9	3	**1, 3, 6, 9, 9**	**28**	**9**
Thu	1	0	6	0	8	**0, 0, 1, 6, 8**	**15**	**0**
Fri	2	4	3	4	9	**2, 3, 4, 4, 9**	**22**	**4**
Sat	1	3	1	3	1	**1, 1, 1, 3, 3**	**9**	**1**
Sunday	1	7	5	7	6	**1, 5, 6, 7, 7**	**26**	**7**
Ratings	**1, 1, 1, 1, 1, 1, 2**	**0, 2, 3, 4, 5, 7, 9**	**1, 1, 3, 5, 6, 6, 6**	**0, 2, 2, 3, 4, 7, 9**	**_, _, _, 5, 6, 8, 9**			
Sum	**8**	**30**	**28**	**27**	**33**			
Mode	**1**	**-**	**6**	**2**	**1**			

For E:

The remaining rating = 33 - (1 + 3 + 8 + 9 + 1 + 6) = 5

Day	**A**	**B**	**C**	**D**	**E**	**Ratings**	**Sum**	**Mode**
Monday	1	2	1	2	1	**1, 1, 1, 2, 2**	**7**	**1**
Tue	1	5	6	2	5	**1, 2, 5, 5, 6**	**19**	**5**
Wed	1	9	6	9	3	**1, 3, 6, 9, 9**	**28**	**9**
Thu	1	0	6	0	8	**0, 0, 1, 6, 8**	**15**	**0**
Fri	2	4	3	4	9	**2, 3, 4, 4, 9**	**22**	**4**
Sat	1	3	1	3	1	**1, 1, 1, 3, 3**	**9**	**1**
Sunday	1	7	5	7	6	**1, 5, 6, 7, 7**	**26**	**7**
Ratings	**1, 1, 1, 1, 1, 1, 2**	**0, 2, 3, 4, 5, 7, 9**	**1, 1, 3, 5, 6, 6, 6**	**0, 2, 2, 3, 4, 7, 9**	**1, 1, 3, 5, 6, 8, 9**			
Sum	**8**	**30**	**28**	**27**	**33**			
Mode	**1**	**-**	**6**	**2**	**1**			

Rating for each Person:

Day	**Girish**	**Vaibhav**	**Shikha**	**Tarun**	**Harshit**	**Sum**

Monday	1	2	1	2	1	**7**
Tue	1	5	6	2	5	**19**
Wed	1	9	6	9	3	**28**
Thu	1	0	6	0	8	**15**
Fri	2	4	3	4	9	**22**
Sat	1	3	1	3	1	**9**
Sunday	1	7	5	7	6	**26**
Sum	**8**	**30**	**28**	**27**	**33**	**126**

Girish receive a Rating of 1 on Wednesday.

Video Solution:

Q27. Text Solution:

Day	Girish	Vaibhav	Shikha	Tarun	Harshit	Sum
Monday	1	2	1	2	1	**7**
Tue	1	5	6	2	5	**19**
Wed	1	9	6	9	3	**28**
Thu	1	0	6	0	8	**15**
Fri	2	4	3	4	9	**22**
Sat	1	3	1	3	1	**9**
Sunday	1	7	5	7	6	**26**
Sum	**8**	**30**	**28**	**27**	**33**	**126**

On Tuesday:

Shikha's Rating = 6

Harshit's Rating = 5

Required Sum = 6 + 5 = 11

Video Solution:

Q28. Text Solution:

Day	Girish	Vaibhav	Shikha	Tarun	Harshit	Sum
Monday	1	2	1	2	1	**7**
Tue	1	5	6	2	5	**19**
Wed	1	9	6	9	3	**28**
Thu	1	0	6	0	8	**15**
Fri	2	4	3	4	9	**22**
Sat	1	3	1	3	1	**9**
Sunday	1	7	5	7	6	**26**
Sum	**8**	**30**	**28**	**27**	**33**	**126**

On Friday, none of the delivery employees receive a rating of 1.

Video Solution:

Q29. Text Solution:

Day	Girish	Vaibhav	Shikha	Tarun	Harshit	Sum
Monday	1	2	1	2	1	**7**
Tue	1	5	6	2	5	**19**
Wed	1	9	6	9	3	**28**
Thu	1	0	6	0	8	**15**
Fri	2	4	3	4	9	**22**
Sat	1	3	1	3	1	**9**
Sunday	1	7	5	7	6	**26**
Sum	**8**	**30**	**28**	**27**	**33**	**126**

On Thursday:

Rating of Employees with code B = 0

Rating of Employees with code D = 0

Required Sum = 0

Video Solution:

Q30. Text Solution:

Day	Girish	Vaibhav	Shikha	Tarun	Harshit	Sum
Monday	1	2	1	2	1	**7**
Tue	1	5	6	2	5	**19**

Wed	1	9	6	9	3	**28**
Thu	1	0	6	0	8	**15**
Fri	2	4	3	4	9	**22**
Sat	1	3	1	3	1	**9**
Sunday	1	7	5	7	6	**26**
Sum	**8**	**30**	**28**	**27**	**33**	**126**

Vaibhav receive a rating of 7 on Sunday.

Video Solution: